Comparing the Energy Requirements for Maple Syrup Production through Boiling and Freezing Methods

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Introduction/Context

Maple syrup production requires the evaporation of large quantities of water from sap to concentrate the sugars. This process is energy intensive, traditionally relying on wood fire evaporators. To reduce energy consumption and improve energy efficiency, the Maple Nation indigenous people in northeastern America have developed an alternative technique, as described in *Braiding Sweetgrass* by Robin Wall Kimmerer, for concentrating the sugars in sap: pouring the sap into long, shallow troughs and allowing the liquid to freeze overnight.[1] The water in the sap freezes, leaving behind a more concentrated sugar solution. This process, repeated over many nights, results in maple syrup. How energy efficient is the freezing method over the boiling method to produce maple syrup? We seek to answer this question by comparing the energy requirements of each method of syrup production.

Maple syrup consists of 52-76% sucrose, 0-10% fructose, and 0-4% glucose.[2] Maple sap is primarily water (95-99% water and 1-5% sucrose).[3] Although there are many other organic compounds present in maple sap and syrup, such as malic and citric acid, which could influence phase transition behavior, the phase transition behavior of sap into syrup can be approximately modelled using the temperature-composition diagram for a binary system of water and sucrose, as shown in Figure 1. Maple sap begins in the syrup region of the diagram with a low weight percent sucrose and at approximately room temperature (20°C). For the boiling method, as the temperature of the mixture increases, the water boils first and two phases, steam and syrup, are present. As the solution boils between 100°C and 122°C and the steam is allowed to escape, the weight percent sucrose in the liquid solution increases and the liquid becomes more viscous, as shown in Figure 2. Once a desired viscosity (and therefore concentration of sugar) has been reached, the solution is allowed to cool back to room temperature. For the freezing method, the temperature is reduced to between 0°C and -13.9°C until a mixture of syrup, ice, and sucrose is attained. The ice is purely water, and so by removing the ice the concentration of sugar is greatly increased. As long as the sugar weight percent is not above 63.6%, when the solution (sans ice) is allowed to warm up to room temperature, the solution will be a more concentrated syrup.

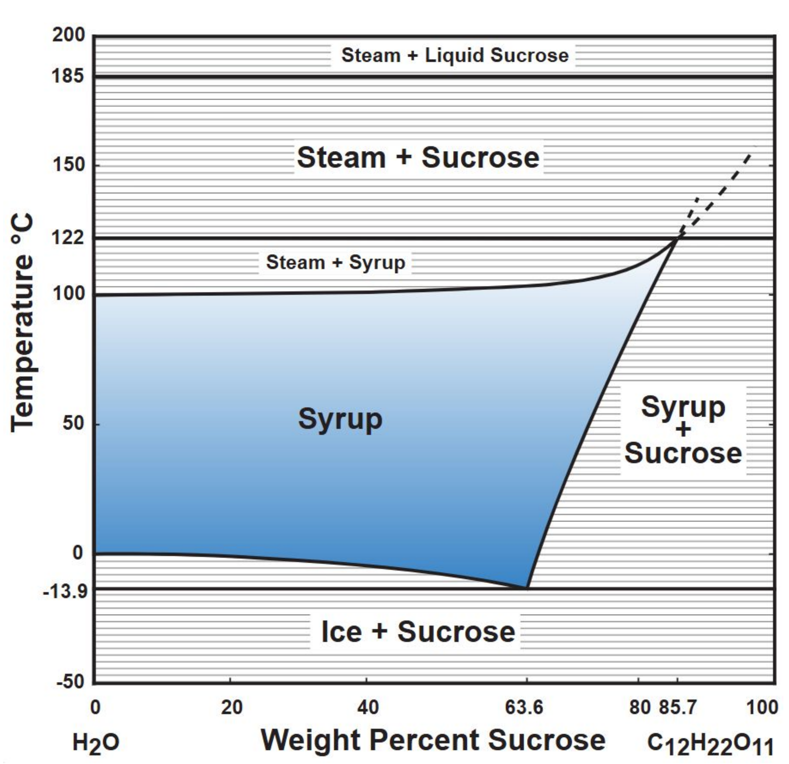

Figure 1—Temperature-Composition Diagram of a Mixture of Water and Sucrose [4]

Figure 1 shows the phases at equilibrium in a binary system of water and sucrose at different temperatures and composition (weight percent sucrose). Pure water boils at 100°C, while sugar water boils at higher temperatures for higher weight percentages of sucrose. The blue area indicates temperatures and compositions where a single liquid phase (syrup) is present.

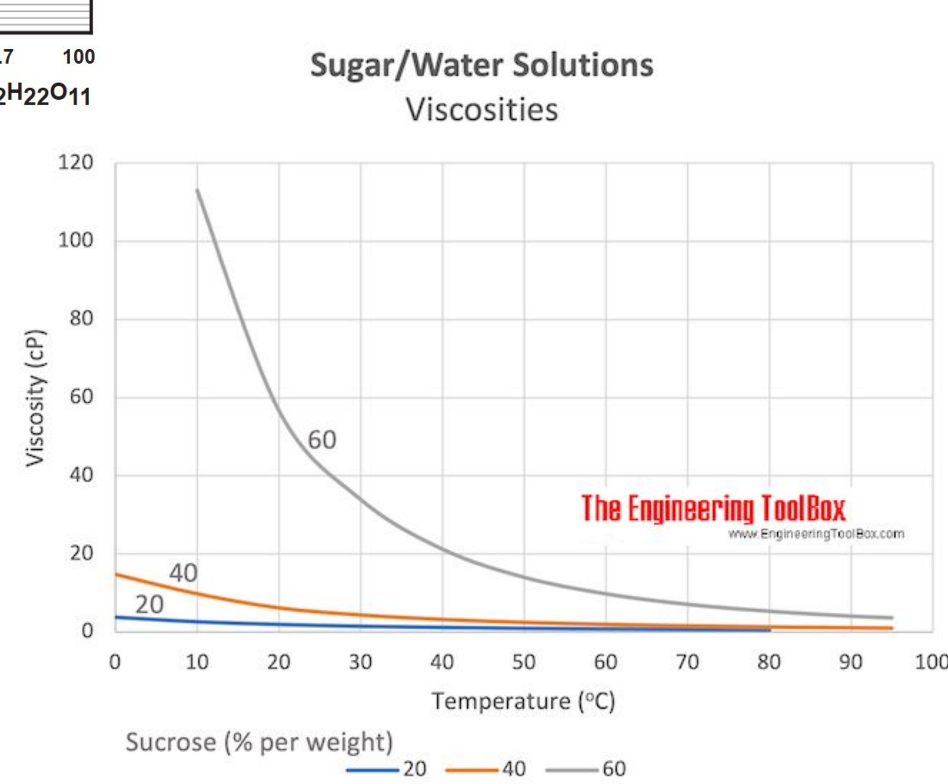

Figure 2— Temperature-Viscosity Diagram of a Mixture of Water and Sucrose [5]

Figure 2 shows the viscosity of a sucrose-water solution as a function of temperature and weight percent sucrose. As the temperature increases, the viscosity decreases. As the concentration of sucrose increases, the viscosity increases.

Methods

Freezing Method: The initial volume of sap was measured using a measuring cup (370 mL). It was frozen in the freezer (-18°C, using R-134a) for about 4 hours, and then the ice was separated from the remaining liquid, resulting in 89 mL of syrup.

Figure 3—Images of Maple Sap Freezing Process

Figure 3 shows the freezing process. On the left, the sap before freezing is shown. In the middle, the sap after freezing. On the right, the produced maple syrup is shown, which is much darker in color than the original sap.

Boiling Method: Roughly 525mL of sap was boiled in a pot, first at high heat to bring the solution to a boil but eventually low to maintain a rolling boil. The viscosity and temperature were closely monitored. After about 40 minutes, the pot was removed from the burner and the syrup was collected. The gas stove used was 17,000 BTU and used over varying settings (1-6) throughout the boiling process.

Figure 4—Images of Maple Sap Boiling Process

Figure 4 shows the sap boiling process. The image on the left shows the sap once it had begun to boil at approximately 100°C. The image in the middle shows the sap after ~35 minutes, and the image on the right shows the produced syrup (7.39 mL). As the solution boiled, the temperature was consistently around 100°C, slowly rising to 106°C and hitting 114°C briefly. The initial syrup was runny, but thickened as the solution cooled.

Calculations and Discussion

Freezing Method: The initial volume of the sap was 370 mL and the final volume was 89 mL. Assuming that the initial sap contained 5% sucrose by weight, the final syrup contained 20% sucrose by weight. From Figure 1, this corresponds to a saturated temperature of -3°C.

Assumptions:

- The sap begins at room temperature (20°C) and reaches -3°C
- The evaporator operates at -18°C and the condenser operates at 48°C
- The evaporator outlet is saturated vapor, the condenser outlet is saturated liquid
- The refrigerator uses R-134a refrigerant
- The valve is well insulated (adiabatic), the compressor is isentropic
- The system experiences no significant change in potential or kinetic energy
- The specific heat capacity of water is 4.18 kJ/kgK,[6] sucrose is 1.24 kJ/kgK,[7] ice is 2.05 kJ/kgK.[8] The enthalpy of fusion of water is 334 kJ/kg.[9] The density of the sap is ~1 g/mL.

Since the valve is well insulated, steady state, and does no work, the first law open system equation simplifies to $\hat{H}_{in} = 120.39\frac{kJ}{kg} = \hat{H}_{out}$ (from saturated R-134a tables)

Therefore, since the valve outlet goes into the evaporator, the evaporator inlet has an enthalpy value of 120.39 kJ/kg. The evaporator outlet enthalpy value is 239.64 kJ/kg (from saturated R-134a tables)

$$\frac{\partial}{\partial t}\left[m\left(\hat{U}+\frac{v^2}{2}+gh\right)\right] = \dot{m}_{in}\left(\hat{H}_{in}+\frac{v^2}{2}+gh\right) - \dot{m}_{out}\left(\hat{H}_{out}+\frac{v^2}{2}+gh\right) + \dot{W} + \dot{Q}$$

$$\dot{Q} = \dot{m}(239.64\frac{kJ}{kg} - 120.39\frac{kJ}{kg})$$

$$\frac{Q}{m} = 119.25\frac{kJ}{kg}$$

So the evaporator removes heat from the sap at a rate of 119.25 kJ/kg.

Assuming that the sap is composed of 95% water and 5% sucrose, the specific heat capacity of the liquid sap can be approximated as c_l = (0.95)(4.18 kJ/kgK) + (0.05)(1.24 kJ/kgK) = 4.033 kJ/kgK. The specific heat capacity of the solid sap can be similarly approximated as c_s = 2.01 kJ/kgK. The total amount of heat required to freeze the sap to -3°C is:

$$\hat{q}_{tot} = \hat{q}_{20°C-0°C} + \hat{q}_{fus} + \hat{q}_{0°C--3°C} = c_l(0°C - 20°C) + \Delta H_{fus} + c_s(-3°C - 0°C)$$

$$= 4.033\frac{kJ}{kgK}(-20°C) - 334\frac{kJ}{kg} + 2.01\frac{kJ}{kgK}(-3°C)m = -420.69\frac{kJ}{kg}$$

So, 420.69 kJ/kg must be removed from the sap. Since the freezing process occurred over 4 hours, the mass flow rate of the refrigerant is

$$\dot{m}\frac{Q}{m} = q_{tot}(0.370kg)\frac{1}{4hr} \Rightarrow \dot{m} = 0.326\frac{kg}{hr}$$

Assuming that the compressor is isentropic, the first law for open systems simplifies to

$$\hat{W} = \hat{H}_{in} - \hat{H}_{in}$$

H_{in} is the enthalpy of the saturated vapor from the evaporator at -18°C, so H_{in}=239.64 kJ/kg. S_{in} = 0.94389 kJ/kg. Since $S_{in}=S_{out}$ and the compressor outlet is a superheated vapor at ~1.2 MPa, linear interpolation was used to find H_{out} = 289.90 kJ/kg. So, the compressor does 44.26 kJ/kg of work. Therefore, (44.26 kJ/kg)(4 hr)(0.326 kg/hr) = **57.72 kJ of energy is required for the freezing process.**

Boiling Method: The initial volume of the sap was 525 mL, the final volume was 7.39 mL.

Assumptions:

- The sap begins at room temperature (20°C) and reaches 106°C
- The density of the sap is ~1 g/mL and is constant
- c_l = 4.033 kJ/kgK (from above calculations)
- The enthalpy of vaporization of water is 2257.06 kJ/kg,[10] the specific heat capacity of steam is 1.996 kJ/kgK [11]

$$0.525kg - 0.00739kg = 0.51761kg \text{ of water evaporated} = m_{vap}$$
$$m_{remain} = 0.00739kg$$

$$q_{tot} = q_{20°C-100°C} + q_{vap} + q_{100°C-106°C}$$

$$= m_l c_l(100°C - 20°C) + m_{vap}(2257.06\frac{kJ}{kg}) + m_{remain}c_{steam}(106°C - 100°C)$$

$$= (0.525kg)(4.033\frac{kJ}{kgK})(80°C) + (0.51761kg)(2257.06\frac{kJ}{kg}) + (0.00739kg)(1.996\frac{kJ}{kgK})(6°C)$$

$$= 1337.8kJ$$

Therefore, 1337.8 kJ of energy is required for the boiling process.

The boiling method required more than 20 times more energy than the freezing method to produce sap. However, it is important to note that both processes did not result in the same final concentration of sucrose in the syrups. The syrup produced through boiling (containing 0.02625 kg sucrose in 7.39 mL) was much more concentrated than the syrup produced through freezing (containing 0.0185 kg sucrose in 89 mL). The syrup produced through boiling was approximately 17 times more concentrated that the syrup produced through freezing. However, since 17 is still less than 20, the boiling method was less energy efficient than the freezing method.

References

1. Kimmerer, R. W. Maple Sugar Moon. *Braiding Sweetgrass: Indigenous Wisdom, Scientific Knowledge, and the Teachings of Plants*; pp 63-71.
2. Stuckel, J. G.; Low, N. H. The Chemical Composition of 80 Pure Maple Syrup Samples Produced in North America. *Food Research International* **1996**, *29* (3–4), 373–379. https://doi.org/10.1016/0963-9969(96)00000-2.
3. Taylor, F. H. *Variation in Sugar Content of Maple Sap*, **1956**. https://www.uvm.edu/d10-files/documents/2024-10/sapsugarcontentvariation.pdf.
4. *Binary Phase Diagrams.* https://www.science.smith.edu/~jbrady/petrology/igrocks-diagrams/binary/ice-sucrose-steam.php
5. *Sugar Water Solutions - Viscosities.* https://www.engineeringtoolbox.com/sugar-solutions-dynamic-viscosity-d_1895.html (accessed 2025-04-28).
6. *Specific Heat Capacity of Water: Temperature-Dependent Data and Calculator.* https://www.engineeringtoolbox.com/specific-heat-capacity-water-d_660.html (accessed 2025-04-28).
7. Informatics, N. O. of D. and. *Sucrose.* https://webbook.nist.gov/cgi/cbook.cgi?ID=C57501&Mask=2 (accessed 2025-04-28).
8. *Ice - Thermal Properties.* https://www.engineeringtoolbox.com/ice-thermal-properties-d_576.html (accessed 2025-04-28).
9. *10.10: Enthalpy of Fusion and Enthalpy of Vaporization.* Chemistry LibreTexts. https://chem.libretexts.org/Bookshelves/General_Chemistry/ChemPRIME_(Moore_et_al.)/10%3A_Solids_Liquids_and_Solutions/10.10%3A_Enthalpy_of_Fusion_and_Enthalpy_of_Vaporization (accessed 2025-04-28).
10. *Phase changes.* http://wpage.unina.it/petrilli/gasvapor/steam.htm?html (accessed 2025-04-28).
11. *Thermal Properties of Water.* https://www.engineeringtoolbox.com/water-thermal-properties-d_162.html (accessed 2025-04-28).